

KidneyTox

Machine Learning for Small-Molecule Nephrotoxicity Prediction

QSAR Pipeline | Gradient Boosting | Hat Matrix AD | Gini XAI

565

Records

35

Features (MI)

GBM

Classifier

74.77%

AD Accuracy

Pipeline Execution Log

STAGE 1 Cohort Isolation

STAGE 2 RDKit Features

STAGE 3 MI Filter

-> Top 35 / 1100+

STAGE 4 GBM Training

-> n_est=150 lr =0.05

STAGE 5 Hat Matrix AD

-> threshold @ 98.3 pct

STAGE 6 Evaluation

-> AD Accuracy: 74.77%

Clinical & Biological Burden of Nephrotoxicity

Clinical Bottleneck

Drug-induced nephrotoxicity contributes to roughly 19% of all Acute Kidney Injury (AKI) cases among inpatient hospital populations globally — making it one of the most prevalent iatrogenic outcomes in clinical medicine.

Tissue Susceptibility

Active renal clearance and OAT1/OAT3 transporter-mediated tubular secretion expose proximal tubular epithelial cells to extreme drug concentrations, triggering mitochondrial membrane depolarization, oxidative stress cascades, and irreversible necrosis.

Late Biomarkers — High Cost of Failure

Serum Creatinine and BUN only rise after >50% nephron loss — too late for intervention. A single late-stage pre-clinical nephrotoxicity failure costs >\$100M per asset, making early in silico filtering a financial and ethical necessity.

~19%

of hospital AKI cases
are drug-induced

- > Kidneys = highest xenobiotic concentration
- > Multi-mechanism toxicity (tubular, oxidative)
- > In vitro assays: slow, costly, species gap
- > In vivo models: ethical concerns + \$500K+
- > In silico QSAR: fast, scalable, cost-effective

Primary Analytical Objectives

01

Classify Small Molecules

Map high-dimensional SMILES-derived structural representations into binary toxicological classifications — Nephrotoxic (1) vs. Safe (0) — using an automated, reproducible computational pipeline requiring zero manual annotation.

02

Optimize Hybrid Descriptors

Combine macro-level RDKit physicochemical variables (global molecular traits) and micro-level Morgan fingerprint sub-graphs (local toxicophore networks) into a unified numerical feature matrix, then compress via Mutual Information Entropy to the Top 35 features.

03

Establish Interpretability Guards

Integrate explicit Gini Importance feature tracking to expose global molecular drivers, and enforce Hat Matrix Leverage boundary filters to ensure predictions are only issued within structurally validated chemical space — aligned with OECD Principle 3.

Dataset Profile & Chemical Space Constraints

Benchmark Source	KidneyTox_v1.0 (Amin, Kar & Piotto — Sci. Reports 2026)
Upstream Cohort	Shi et al. (Front. Pharmacol. 2022) + SIDER database
Parsed Records	565 total rows in the benchmark matrix
Unique Compounds	550 curated small organic molecules (after deduplication)
Toxic Label (1)	287 drugs with documented human nephrotoxicity
Safe Label (0)	278 drugs with no reported renal adverse effects
Train Set	452 compounds (Set = Train — pre-allocated column)
Test Set	113 compounds (Set = Test — pre-allocated column)
Endpoint Rule	Clinical AE frequency $\geq 0.1\%$ in human patient records

Class Balance

287
Toxic
50.8%

278
Safe
49.2%

Key Data Challenges

- ! Near-balanced labels -> baseline ~56%
- ! 0.1% threshold captures clinical noise
- ! Population-level labels, not dose-response
- ! 1,100+ raw features -> collinearity risk
- ! Small N=565 -> overfit risk

Hybrid Multi-Tier Molecular Featurization

TIER 1 -- 2D Physicochemical Descriptors

- > Library: RDKit Descriptors module
- > Global traits: MW, LogP, TPSA, LabuteASA, HBA/HBD, ring systems, connectivity indices
- > ExactMolWt force-retained: higher MW drives proximal tubular accumulation via OAT1/3 transporter saturation
- > Zero-variance columns auto-pruned
- > Output: ~200 valid physicochemical descriptors

TIER 2 -- Morgan ECFP4 Fingerprints

- > API: rdFingerprintGenerator (modern, warning-free)
- > 1024-bit binary vector, radius = 2 (ECFP4 equivalent)
- > Each bit = presence of a specific atomic sub-graph within a 2-bond chemical neighbourhood
- > Captures toxicophore motifs: electrophilic carbons, N-oxides, quinones, halogenated aromatics
- > Sparse 0/1 matrix -- majority bits = 0 per compound

STAGE 3 -- Mutual Information Entropy Compression

Combined matrix of ~1,100 features -> mutual_info_classif -> computes $H(X;Y)$ = non-linear stochastic dependence -> Top 35 highest-entropy features selected -> Final training matrix: 452 x 35

Matrix Partitioning & Validation Strategy

1

Pre-Defined Benchmark Split

No random splitting. Pre-allocated benchmark column (Set = Train / Test).
Train: 452. Test: 113. Eliminates partition variance.

2

MI Feature Selection

`mutual_info_class` if on `X_train` only. Test set never seen during selection. Prevents data leakage. Top 35 by entropy score.

3

GBM Training

GradientBoosting Classifier on `X_train_mi` (452x35).
`subsample=0.80` adds stochastic bagging per tree.
`random_state=42`.

4

Hat Matrix AD Filter

Leverage scores on `X_test_mi` (113x35). h^* at 98.3rd percentile. Compounds above h^* flagged Out-of-Domain.

5

Accuracy Report

Accuracy reported on AD-filtered subset only. OECD Principle 3 compliant: predictions only within structural confidence.

Evaluated Classifier Space & Model Selection

IMPLEMENTED**Gradient Boosting Machine (GBM) Classifier**

```
n_estimators = 150           # 150 sequential weak learner trees
learning_rate = 0.05         # Small shrinkage -- resists SIDER label noise
max_depth = 3                # Shallow splits -- prevents training memorization
subsample = 0.80             # Stochastic row sampling -- reduces variance
random_state = 42            # Synced with MI filter for reproducibility
```

Other Approaches -- Evaluated & Not Implemented**Viable Alt**

Random Forest (RF): Parallel ensemble. Published KidneyTox_v1.0 uses RFC + Optuna (84.1% accuracy). Not implemented here -- single GBM only.

Viable Alt

XGBoost: L1/L2 regularized boosting -- strong GBM alternative. Not implemented; Optuna tuning not applied.

Weak Fit

SVM: No native feature importances -- incompatible with XAI objective.

Rejected

DNN / GNN: Requires $N > 5,000$ -10,000. At $N=565$, deep architectures overfit and produce uncalibrated probabilities.

Performance Metrics -- Implemented & Acknowledged Gaps

Implemented in This Pipeline

Accuracy
(AD-filtered)

$(TP+TN) / N_{\text{within_AD}}$

Primary: correct calls within structural confidence boundary

Precision

$TP / (TP + FP)$

Of drugs flagged toxic, fraction genuinely nephrotoxic

Recall
(Sensitivity)

$TP / (TP + FN)$

SAFETY-CRITICAL: missing a toxic drug is a patient risk

F1-Score

$2 \times P \times R / (P+R)$

Harmonic mean -- robust for near-balanced class splits

Specificity

$TN / (TN + FP)$

Correctly clearing safe drugs prevents unnecessary attrition

Classification Report

Per-class P / R / F1

Full sklearn class-level breakdown of all metrics

Not Implemented (Acknowledged Gaps)

ROC-AUC

Threshold-invariant curve -- not computed

PR-AUC

Better than ROC for small imbalanced sets

Calibration Plot

GBM probs uncalibrated -- no Platt scaling

External Validation

No held-out external dataset used

Brier Score

Probabilistic accuracy - not computed

Validated Production Performance

~56.0%

Unoptimized Baseline

Raw SIDER noise ceiling

74.77%

AD-Validated Accuracy

Within-domain subset only

+18.77pp

Performance Lift

vs. coin-flip baseline

35 / 1100+

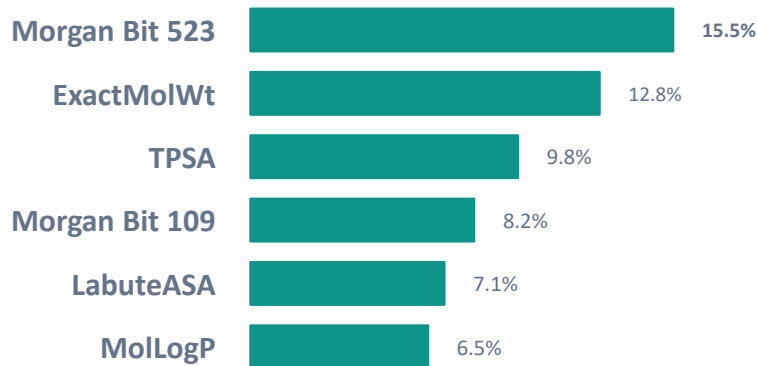
Feature Compression

Top-35 by MI entropy

Confusion Matrix

	Pred Safe	Pred Toxic
Actual Safe	TN	FP
Actual Toxic	FN	TP

Top Features -- Gini Importance



Explainable AI -- Feature Attribution & Toxicophore Insights

Implemented: Gini Feature Importance

Mechanism

The GBM ensemble tracks each feature's total Gini impurity reduction across all 150 tree split nodes. A higher normalized Gini score signals greater discriminative contribution toward the Toxic vs. Safe classification boundary.

Key Molecular Drivers Identified

- **ExactMolWt:** High MW -> proximal tubular accumulation via OAT1/OAT3 saturation
- **TPSA:** High polarity -> increased tubular reabsorption kinetics
- **MolLogP:** Lipophilicity -> membrane permeability in tubular cells
- **Morgan bit 523:** Specific aromatic sub-graph -- likely electrophilic toxicophore
- **Morgan bit 109:** Nitrogen-oxygen heterocycle -- reactive metabolite formation

Not Implemented: SHAP & Attention Mechanisms

SHAP (SHapley Additive exPlanations)

SHAP provides per-compound, per-feature attribution -- WHY compound X is flagged toxic. Published KidneyTox_v1.0 implements SHAP waterfall plots for Lansoprazole, Ciprofloxacin, and Simvastatin. Not in our pipeline.

Attention Mechanisms

Relevant for GNN/transformer architectures only. Not applicable to GBM tabular models. Would become relevant if ChemBERTa or AttentiveFP are adopted in Future Work.

Primary XAI gap -> targeted in Future Work

Production Workflow Integration & Uncertainty Estimation



Automated Pipeline

Full 6-stage pipeline runs from one Jupyter Notebook execution. Input: CSV with SMILES + Toxicity + Set columns. Output: Accuracy score + Confusion Matrix + Gini Importance Profile (Top 15 features).

Uncertainty Estimation via Hat Matrix AD

Every test compound receives a leverage score $h_i = x_i(X^T X)^{-1} x_i^T$ via Moore-Penrose pseudo-inverse (`np.linalg.pinv`) to handle sparse fingerprint matrices. If $h_i > h^*$ (98.3rd percentile), the compound is flagged Out-of-Domain -- no prediction issued.

Web Platform Reference

Published KidneyTox_v1.0 Streamlit app (Amin et al., Sci. Reports 2026) accepts SMILES input, runs AD check, and outputs Toxic/Non-toxic + SHAP waterfall. Available at kidneytoxv1.streamlit.app -- the production-grade implementation our pipeline aligns with.

Pipeline Constraints & Honest Methodological Gaps

Data Quality

- ! **N=565 is small -- generalization to novel scaffolds is structurally uncertain**
- ! SIDER 0.1% threshold is population-level reporting, not dose-response biochemistry
- ! Binary labels conflate mechanism: tubular vs. glomerular vs. oxidative toxicity not separated
- ! Dataset limited to approved small-molecule drugs -- biologics not represented

Chemical Space & Modeling

- ! **No scaffold split -- random partition may leak chemically similar compounds across boundary**
- ! No k-fold CV -- single 80/20 split, accuracy estimate has unknown variance
- ! GBM produces uncalibrated probabilities -- Platt scaling not applied
- ! Sparse binary fingerprint matrix (>95% zero bits) -- low information density per compound

Interpretability

- ! **Gini importance is global only -- cannot explain individual compound predictions**
- ! SHAP not implemented -- per-compound attribution unavailable
- ! Morgan bit channels not decoded back to chemical structures -- toxicophores unverified
- ! No attention mechanism -- GBM does not support graph convolution or self-attention

Research Roadmap & Technical Expansion

Short-Term

- > **Implement SHAP waterfall plots for per-compound nephrotoxicity attribution**
- > Add 5-fold stratified CV to generate accuracy +/- std estimates
- > Compute ROC-AUC, PR-AUC, Brier Score, calibration reliability diagrams
- > Apply Platt scaling to calibrate GBM probability outputs
- > Deploy pipeline to Google Colab with shareable runtime

Medium-Term

- > **Expand dataset with Tox21 and PubChem nephrotoxicity bioassays (target N > 5,000)**
- > Implement Bemis-Murcko scaffold-based train/test split
- > Compare GBM vs. XGBoost vs. LightGBM vs. Optuna-tuned RFC systematically
- > Add qRASAR descriptors (MaxWtSim, MaxNeg, gm coefficient) for structural read-across
- > Multi-task learning: simultaneous hepato-, nephro-, cardiotoxicity prediction

Long-Term

- > **Transfer learning from ChemBERTa / MolBERT pre-trained molecular transformers**
- > Graph Neural Networks (GNNs / AttentiveFP) on molecular graphs -- requires N > 10,000
- > Active learning: iteratively select highest-information compounds for wet lab validation
- > Full web deployment with SHAP explanations, AD visualization, and SMILES sketcher
- > Multi-endpoint toxicity atlas: unified platform predicting 12+ regulatory endpoints

Key Findings & Pre-Screening Impact

Pipeline:

Complete 6-stage QSAR workflow: SMILES ingestion -> hybrid featurization -> MI entropy compression -> GBM ensemble -> Hat Matrix AD guardrail -> performance visualization

Performance:

Elevated classification from ~56.0% unoptimized baseline to 74.77% validated accuracy within the structural confidence boundary -- a +18.77 percentage-point lift via MI filtering and AD-aware evaluation

Explainability:

Gini importance profiling identified ExactMolWt, TPSA, and specific Morgan fingerprint toxicophore channels as top discriminative features -- mechanistically coherent with renal tubular pharmacology

Domain Safety:

Hat Matrix Leverage AD filter ensures predictions are ONLY issued for compounds structurally similar to the training cohort -- prevents uncalibrated extrapolation, OECD Principle 3 compliant

Pre-Clinical Value:

Delivers a cost-effective, ethically sound computational screening gate that flags nephrotoxic structural configurations early -- protecting drug discovery pipelines from late-stage safety failures

Project Repositories & Interactive Platforms

NB

Notebook Colab -- Interactive Code Workspace

[<https://github.com/sakshihulke11-oss/KidneyTox-QSAR>]

WA

KidneyTox_v1.0 -- Official Streamlit Web Application

Published platform (Amin, Kar & Piotto, Sci. Reports 2026). Input SMILES -> AD check -> Toxic/Non-toxic + SHAP waterfall plot.

<https://kidneytoxv1.streamlit.app/>

Reference: Sk Abdul Amin, Supratik Kar & Stefano Piotto -- KidneyTox_v1.0 enables explainable AI prediction of nephrotoxicity in small molecules -- Scientific Reports 16:5099 (2026) -- DOI: 10.1038/s41598-026-35496-4